



State of health estimation of lithium-ion batteries based on fine-tuning or rebuilding transfer learning strategies combined with new features mining

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ABSTRACT

Accurate state of health (SOH) estimation of lithium-ion batteries is essential to ensure the reliability of power equipment. However, the degradation trajectory of different cells and different types of batteries is not repeatable. At present, there is no unified model or method to effectively predict SOH for all batteries. Therefore, a new SOH estimation method is proposed in the paper. Firstly, two types of new features are proposed in this paper. One is the voltage features extracted from the constant-current charging stage, and the other is the capacity recovery feature. They are used to reflect the nonlinear degradation process of the battery. Secondly, the relationship between features and SOH is established by using the LSTM model, which can prevent the problem of gradient vanishing and gradient explosion during model learning. Finally, for the inconsistencies between the same type or different types of batteries, two different transfer learning strategies (fine-tuning and rebuilding) are proposed in this paper, and the effectiveness of the proposed features and transfer learning strategies is verified on three open-source battery data sets (NASA, Oxford, and CALCE). Experimental results show that the SOH estimation method proposed in the paper has good universality, robustness, and accuracy.

1. Introduction

The continuous use of fossil fuels makes the problem of environmental pollution increasingly serious, and the development of environmentally friendly and efficient clean energy equipment is imminent [1,2]. Lithium-ion batteries (LIBs) have been widely used in various fields such as new energy vehicles and energy storage due to the advantages of high power density, superior stability, and long lifetime [3,4]. Compared to hydrogen combustion systems [5,6] and proton exchange membrane fuel cells [7,8], LIBs are low-cost to prepare, convenient to reuse, and more cost-efficient. However, in the process of use, the environmental factors and the electrochemical reaction inside the batteries will lead to irreversible degradation [9], and such degradation will influence the performance of batteries and even cause a safety risk [10,11]. Therefore, in order to ensure the normal operation of the equipment and avoid the occurrence of some safety accidents, it is necessary to accurately estimate the State of Health (SOH) of battery which can indicates the aging of it.

Battery capacity degradation can be attributed to the loss of lithium-ion materials and active materials [12]. This is mainly caused by

physical and chemical side reactions inside the cell, such as graphite shedding, electrolyte loss, solid electrolyte interface (SEI) film formation and continuous thickening, lithium plating, and so on. From the macroscopic scale, the aging of the battery is manifested as the degradation of battery capacity and the increase of internal resistance [13–15]. In practical applications, the ratio of the current available capacity to the rated capacity is often used to indicate the SOH value of the battery.

At present, the methods for SOH estimation can be categorized into the direct measurement method, model-based method, and data-driven method [16,17]. As shown in Fig. 1.

The direct measurement method is mainly to obtain the aging parameters of the battery directly through the experimental equipment and then estimate the SOH of the battery. For example, hybrid pulse power characterization (HPPC) experiments or electrochemical impedance spectroscopy (EIS) [18,19] are used to obtain the internal resistance or impedance information directly to estimate the SOH of the battery, but the measurement of internal resistance or impedance requires special experimental equipment, and the accuracy of the estimate is affected by the experimental equipment. The coulomb counting

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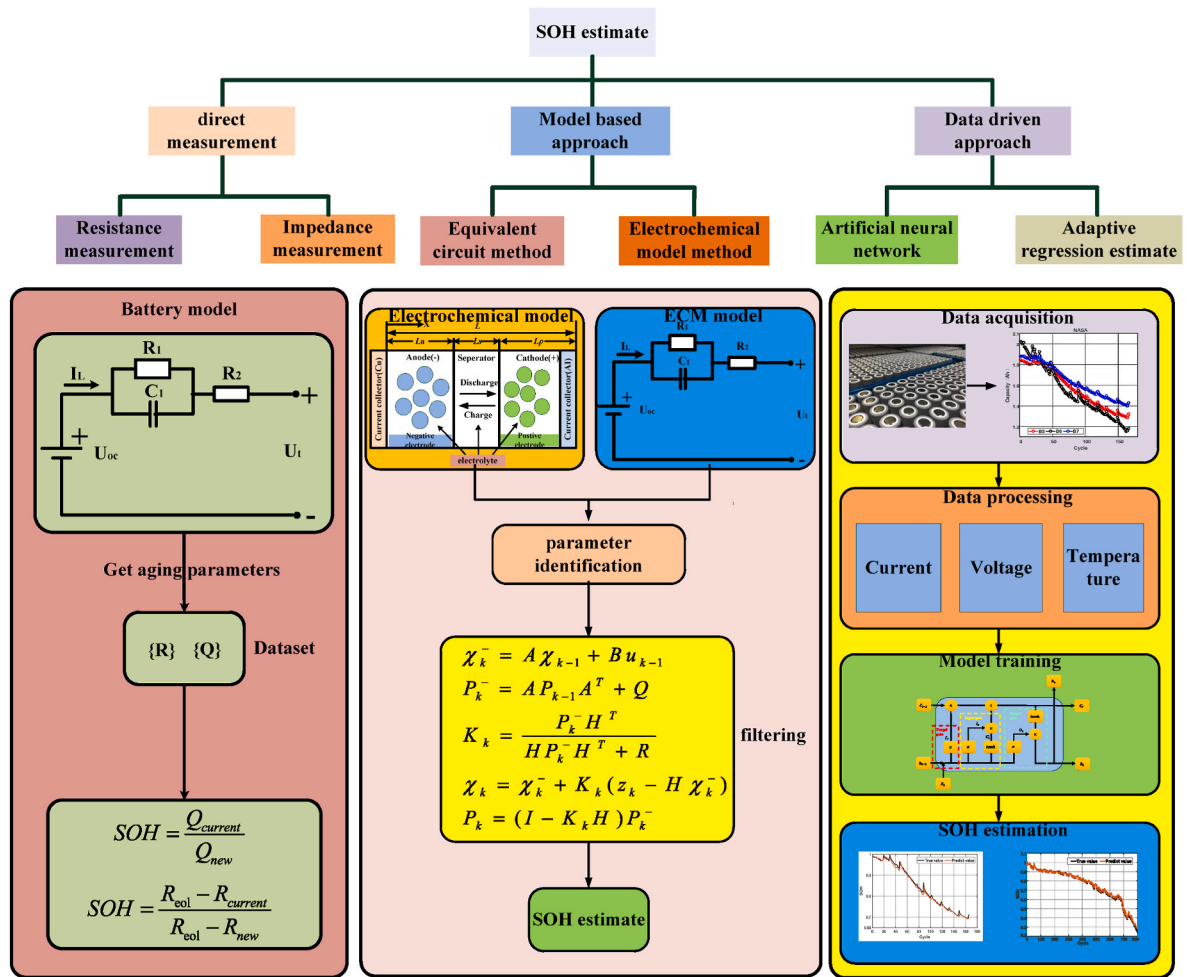


Fig. 1. SOH estimation methods.

method is used to integrate the current of the charging and discharging process to obtain the SOH, which realized an accurate estimation of SOH [20,21]. However, this method requires complete charging and discharging the battery, which is difficult to meet in practice and is time-consuming and labor-intensive [22].

The model-based method focuses on building a dynamic model that corresponds to the battery degradation process, which mainly includes the electrochemical model (EM) and equivalent circuit model (ECM). Then the SOH of the battery is estimated by updating the model parameters through methods such as online identification and filtering [23–26]. Xu et al. [27] used Thevenin equivalent circuit model and developed a novel adaptive dual extended Kalman filter (ADEKF) algorithm. Experimental results show that this algorithm can accurately track the changes of ohm internal resistance and has good convergence. To verify the universality of the method, Ling et al. [24] proposed a SOH estimation method based on a double-extended Kalman filter (EKF) and fractional-order model. Experimental results show that the dual filter not only obtains more accurate SOC and model voltage for accurate SOH

estimation but also has good stability and robustness under three different experimental tests. Bi et al. [28] and Li et al. [29] proposed a single-particle degradation model, which has a simple structure and realizes the rapid and accurate prediction of SOH. However, the estimation accuracy of the model-based estimation method depends on the model's accuracy.

In contrast to the model-based approach, the data-driven method does not require modeling the physical or chemical reactions inside the battery, but mines the hidden health information from the degradation data of the battery. Then combines deep learning algorithms to realize the SOH estimation [30]. The data-driven approach works on the equivalent of a black-box model, which needs to combine data to learn the correspondence between input and output through training [31]. Its main processes can be summarized into the following four steps, as shown in Fig. 2.

- (1). Obtain the battery data, including current, voltage, temperature, and other data.

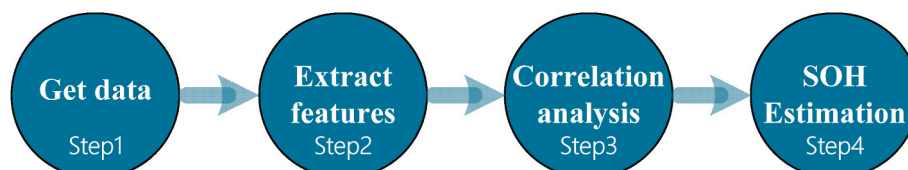


Fig. 2. Main processes of the data-driven method.

Table 1

The advantages and disadvantages of the three methods.

Method	Advantage	Disadvantage
Direct measurement method	Low computational complexity	The accuracy of the measurement is affected by the experimental equipment
Model-based method	Strong robustness, versatility, and high accuracy	The accuracy of the estimation results depends on the accuracy of the battery model
Data-driven method	Strong robustness, versatility, and high accuracy	The accuracy of the prediction is limited by the amount of training data

- (2). Extract valid features from the obtained battery data.
- (3). Analyze the correlation degree between extracted features and SOH with correlation analysis.
- (4). The features with strong correlation can be directly used for training and prediction; The features with weak correlation can be optimized and then reanalyzed, or discarded the weak features and then extract other features.

At present, lots of data-driven methods have been developed, including artificial neural network (ANN) [32,33], extreme learning machine(ELM) [34,35], support vector machine and support vector regression(SVR) [36–39], gaussian process regression(GPR) [40,41], and so on. Cheng et al. [42] combined the Empirical Pattern decomposition (EMD) method with the backpropagation of long short-term memory neural network (B-LSTM) to estimate SOH. The EMD method and the many-to-one B-LSTM neural network structure can effectively reduce the interference of other factors and improve the accuracy of SOH estimation. To improve the accuracy and robustness of SOH estimation, Xiong et al. [43] proposed an online SOH estimation method based on the weighted least squares support vector Machine (WLS-SVM). In the literature, health feature factors (HFs) were extracted from the partial charging curve of batteries, and Pearson correlation coefficient analysis was used to select health indicators with a high correlation with SOH. Finally, the accuracy of SOH estimation is improved by improving the weighting function and linear equations. Chang et al. [44] proposed an online method based on incremental capacity (IC) and wavelet neural networks to estimate SOH, which optimized initial connection weights, translation factor, and the scaling factor of WNN by genetic algorithm to improve the accuracy of SOH estimation. Sui et al. [45] proposed battery voltage fuzzy entropy (FE) as a new feature of SOH estimation. The influence of temperature change on battery degradation was considered in the SOH estimator established, and the method was compared and analyzed with the traditional sample entropy estimation method. Finally, the experimental results verified the effectiveness of the method. Dai et al. [46] developed an a priori knowledge neural network and Markov chain to predict the SOH of a single lithium-ion battery. The extracted features can effectively capture the process of cell degradation and improve the accuracy of SOH estimation. However, this approach requires a large amount of data to train the model, which is difficult to satisfy in practical applications. Li et al. [47] developed a method to predict SOH through a variable long short-term memory neural network (AST-LSTM NN), which determines old information and new data simultaneously through a fixed connection. The element-wise product of new inputs and historical cell states are conducted for screening out more beneficial information. Experiments on NASA datasets proved the effectiveness and accuracy of the method. However, the data-driven method relies on large amounts of data for model training.

The advantages and disadvantages of the three methods are shown in Table 1. Among the three methods, the data-driven method will be the future trend because it does not require too much modeling and complex proof [39].

In conclusion, many achievements have been made in SOH

estimation methods. However, due to the complex electrochemical characteristics of batteries, there still exist the following challenges: (1) The available capacity degradation of the battery is not monotonically decreasing, when the battery has rested for a long time or discharged with a small current, it will cause a brief recovery of the available capacity which called capacity recovery phenomenon. It often hurts the estimation accuracy [48,49]. (2) Due to the long life of batteries, obtaining the training data needed for modeling is a time-consuming process. To use a small amount of data to establish the SOH estimation model, transfer learning has been gradually introduced into the field of batteries in recent years [50], where the degradation process of existing batteries can be transferred to other batteries. However, only when the training data and the test data have the same or similar distribution, the model can obtain high estimation accuracy [51]. The inherent inconsistency of batteries makes even the same type of batteries have different degradation patterns. As a result, on the one hand, the model established by one battery is difficult to apply to other types of batteries, on the other hand, even with the same types of batteries, it is possible to obtain unsatisfied estimation accuracy.

In view of the abovementioned issues, a new SOH estimation framework is put forward in the paper, and the main contributions are as follows.

- (1) To characterize the degradation state of batteries effectively, a new combined health indicator is proposed in the paper. It extracts seven health features from two aspects. One aspect is to describe the shape and the length of the charging curves which change regularly with the aging of the battery, the other aspect is to reflect the degree of the capacity recovery. The experimental results show that the health features proposed in this paper could get SOH results with high precision.
- (2) The battery degradation process is complex and nonlinear. To reflect the relationship between health features and SOH and the temporal correlation of the degradation process, a neural network structure combining LSTM with a dropout layer and three full connection layers is established in the paper. Three open-source data are used to verify the performance of the model and the results prove its effectiveness and universality.
- (3) For transfer learning, considering that different types of batteries have strong inconsistency while the same type of batteries has relatively weak inconsistency, two different transfer learning strategies of rebuilding and fine-tuning are proposed in this paper. In which, the rebuilding strategy mainly adjusts the parameters of the LSTM layer and generates new fully connected layers, the fine-tuning strategy only adjusts the parameters of the fully connected layers. The experimental results demonstrate the effectiveness of the two strategies.

The remainder of the paper is organized as follows: In Section 2, the battery dataset used in this paper is explained; Section 3 introduces SOH estimation methods, including feature extraction, model building, and transfer learning. In Section 4, the results of SOH estimation under different conditions are analyzed and discussed. Finally, the conclusion is drawn in Section 5.

2. Experimental dataset

To facilitate the description of lithium-ion battery characteristics, the battery datasets will be introduced first. In addition, to verify the effectiveness and generality of the method proposed in this paper, three open-source datasets will be used, they come from NASA Prognostics Center of Excellence (PCoE) [52], University of Oxford [53], and the Center for Advanced Life Cycle Engineering (CALCE) [54,55] respectively.

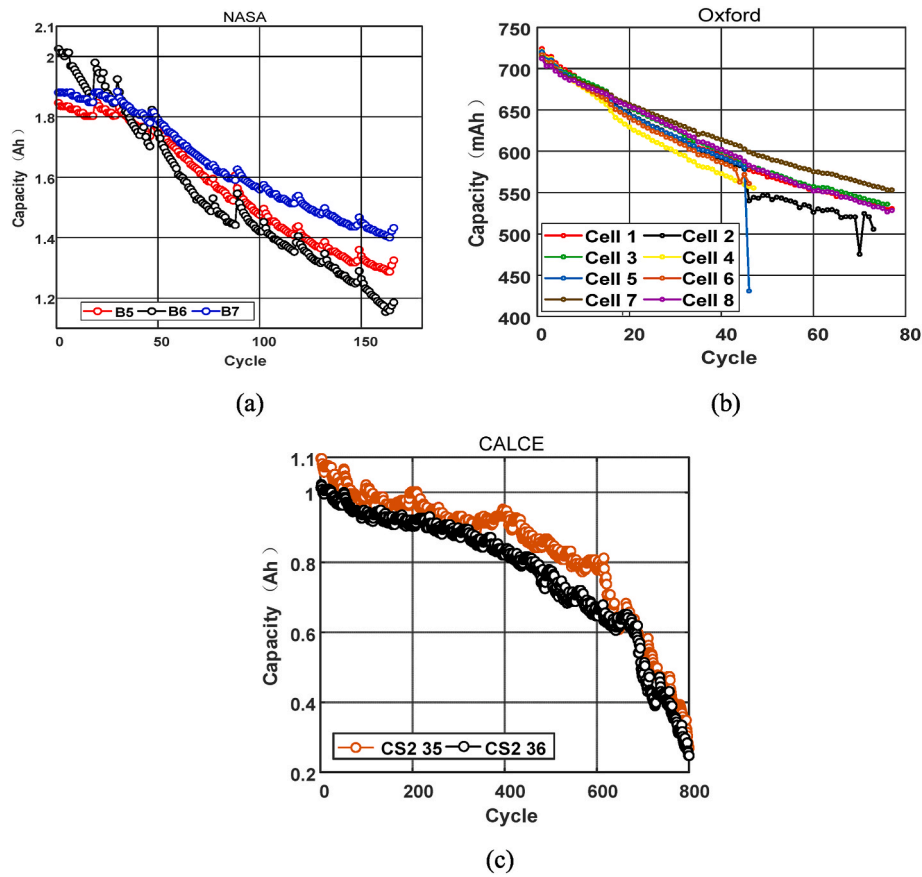


Fig. 3. (a) NASA battery capacity degradation curve. (b) Oxford battery capacity degradation curve. (c) CALCE battery capacity degradation curve.

2.1. NASA dataset

The NASA laboratory used 18650 lithium-ion battery, carried out the cyclic charge-discharge aging experiment at room temperature of 24 °C, and obtained data such as current, voltage, and temperature. The B0005, B0006, and B0007 battery datasets were selected for this article. For convenience of expression, B0005, B0006, and B0007 are denoted as B5, B6, and B7 in this paper. In the charging, the battery was charged in a constant current (CC) mode at 1.5A until the battery voltage reached 4.2V and then continued in a constant voltage (CV) mode until the charge current dropped to 20 mA. After charging is completed, the discharging was performed at a constant current (CC) mode of 2A until the battery voltage dropped to 2.7V, 2.5V, and 2.2V respectively. NASA battery capacity degradation curve is shown in Fig. 3(a).

2.2. Oxford dataset

The Oxford battery dataset records the battery aging experiment dataset produced by the University of Oxford. The eight batteries labeled Cell 1 to Cell 8 were selected for this paper, with a rated capacity of 740mAh. During the aging experiment, all eight batteries were charged in the constant current mode of 1.48A and discharged under the ARTEMIS condition to simulate the actual driving state of electric vehicles. Measure the degree of battery aging at the current capacity every 100 aging cycles. The voltage, current, and surface temperature of the cells were measured at a temperature of 40 °C with a Bio-LogicMPG-205 tester at 1s intervals. Oxford battery capacity degradation curve is shown in Fig. 3(b).

2.3. CALCE dataset

The CALCE datasets were tested cyclically by the US Arbin BT2000 battery test system. The CS2-35 and CS2-36 datasets were selected for this article. The rated capacity of the battery was 1.1Ah. During the aging experiment, the battery was charged at a constant current(CC) rate of 0.5C until the voltage reached 4.2V, and then it was charged at a constant voltage(CV) of 4.2V until the charging current dropped below 0.05A. After fully charged, the battery was discharged with 1C until the terminal voltage dropped to 2.7 V. CALCE battery capacity degradation curve is shown in Fig. 3(c).

3. Methodology

A SOH estimation framework is put forward in this paper. The procedure of the proposed framework is shown in Fig. 4.

It contains four steps.

Step 1. Data Collection. The method proposed in the paper mainly uses measurable voltage, current and temperature data of the batteries.

Step 2. Feature Extraction. Two types of health features are extracted, one is used to characterize the regular change of the charging voltage curve with battery aging, and the other is used to characterize the capacity recovery.

Step 3. Model Training. First, using the source domain batteries to train a neural network model consisting of LSTM, Dropout, and fully connected layers, which is used as the base model. Then, using the top 25% of the target battery data to retrain the base model so as to obtain the final SOH estimation model for the target battery.

Step 4. SOH estimation. Using the model established in step 3 to

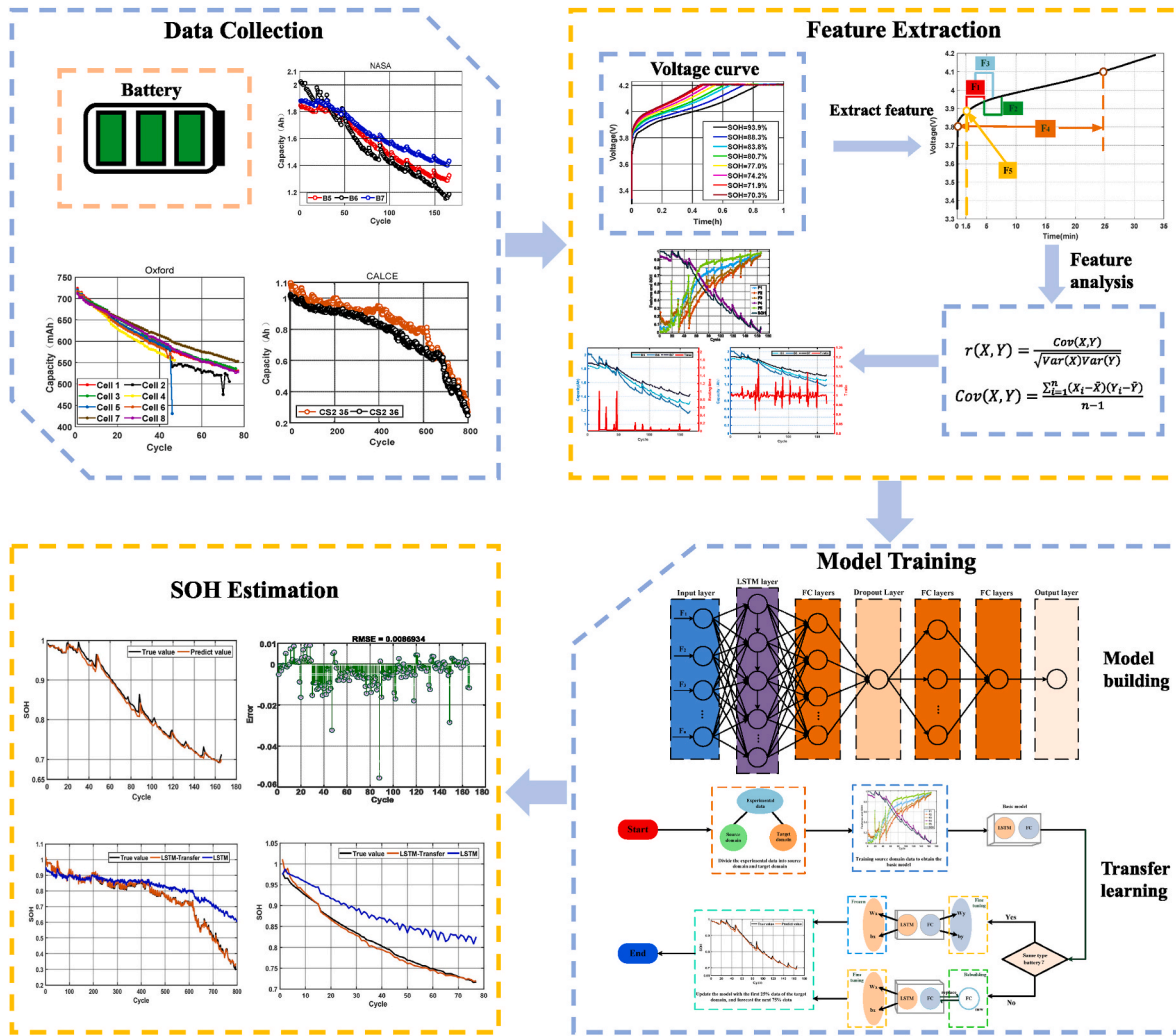


Fig. 4. Overall framework of the method.

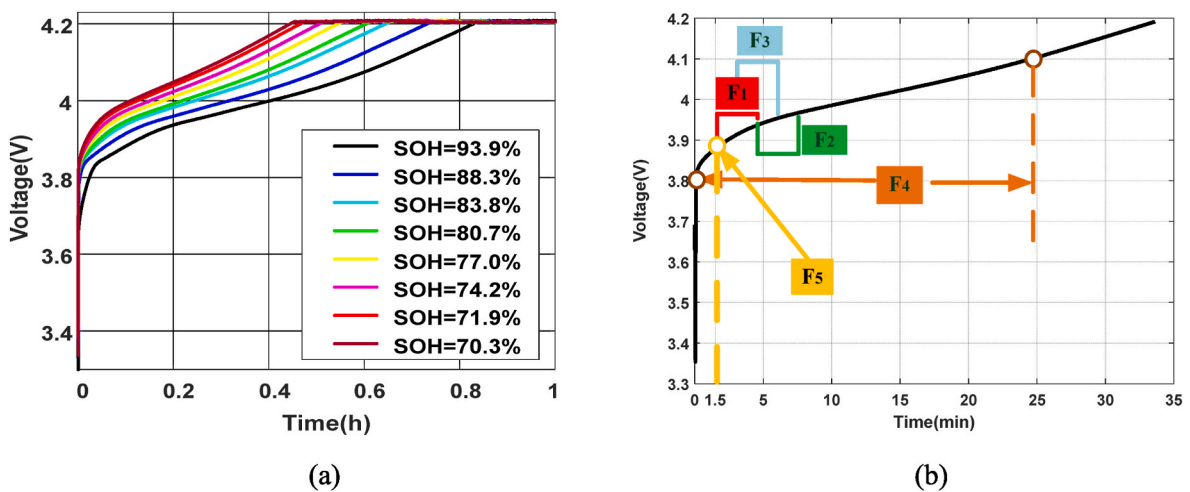


Fig. 5. (a) B5 battery charging voltage curves. (b) Five charging voltage curve features.

estimate the SOH of the target battery.

3.1. Feature extraction and analysis

3.1.1. Feature extraction based on CC charging curve

In practical applications, the discharge process is random and not

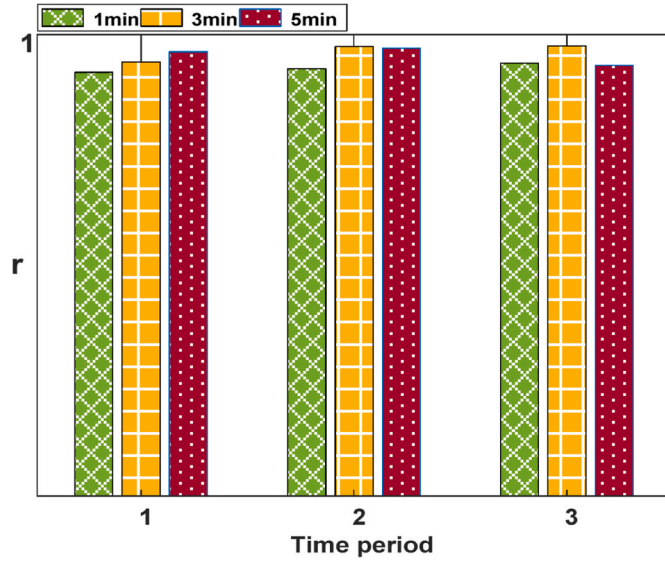


Fig. 6. Correlation analysis.

easy to control, while the charging process is relatively stable. Therefore, this paper chooses to extract health features from the battery charging data.

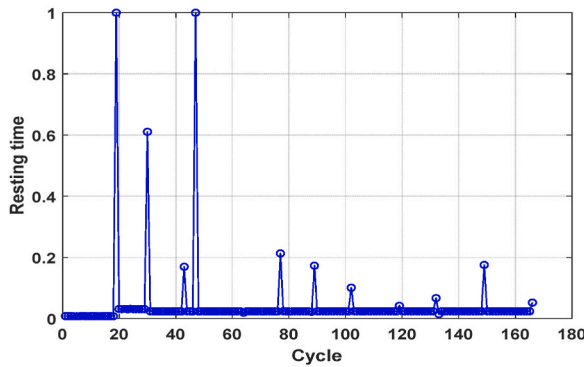
Fig. 5(a) shows the charging voltage curves under different aging states. It can be seen from Fig. 5(a), with the battery degradation, the charging curve changes regularly, that is, the constant current charging time is getting shorter, and the voltage rises faster and faster during the same time. Therefore, health features will be mined based on the voltage curve shape.

To be more specific, it can also be seen that the initial charging of the battery is accompanied by a short period of sudden voltage increase, during which the data fluctuates greatly. Hence, to obtain the stationarity of data, this paper takes 1.5 min of battery charging as the benchmark and then extracts three average voltage features over a short period ($F_1 \sim F_3$), as well as the charging time from 3.8V to 4.1V of the battery (F_4) and the battery voltage at 1.5 min of battery constant current charging (F_5), as shown in Fig. 5(b).

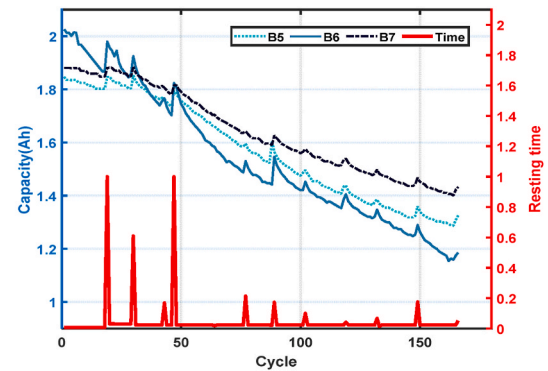
The calculation formula for extracting features is as follows:

$$F_1 = \frac{1}{t_2 - t_1} \left(\int_{t_1}^{t_2} V dt \right) \quad (1)$$

$$F_2 = \frac{1}{t_4 - t_3} \left(\int_{t_3}^{t_4} V dt \right) \quad (2)$$



(a)



(b)

Fig. 7. (a) Resting time of different cycles. (b) Comparison of resting time and SOH.

$$F_3 = \frac{1}{t_6 - t_5} \left(\int_{t_5}^{t_6} V dt \right) \quad (3)$$

where $[t_1, t_2]$, $[t_3, t_4]$, $[t_5, t_6]$ represent three different short periods on the voltage curve, respectively.

$$F_4 = t_{V_a} - t_{V_b} \quad (4)$$

In the formula, t_{V_a} and t_{V_b} respectively indicate the battery charging time when the battery charging voltage rises to 3.8V and 4.1V.

In Fig. 5(b), the short periods of the first three features ($F_1 \sim F_3$) use the same length. Based on the length of the short period, this paper makes three choices: 1 min, 3 min, and 5 min respectively. The Pearson correlation analysis method was used to explore which time period features were more strongly correlated with SOH. The calculation formula of the Pearson correlation coefficient is as follows:

$$r(X, Y) = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)\text{Var}(Y)}} \quad (5)$$

$$\text{Var}(X) = \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n - 1} \quad (6)$$

$$\text{Cov}(X, Y) = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{n - 1} \quad (7)$$

where r represents the correlation coefficient. It can reflect the closeness of the correlation between two variables X and Y , and also reflect whether the change direction of the two variables is the same. r ranges from $[-1, 1]$. The closer the absolute value of r is to 1, the stronger the correlation, while it approaches 0, the weaker the correlation. A positive value indicates a positive correlation, that is, the Y increases as X increases, and a negative value indicates a negative correlation, which means the Y decreases as X increases. $\text{Var}(X)$ represents the variance of X . $\text{Cov}(X, Y)$ represents the covariance between X and Y . n represents the total sample number. X_i and Y_i represent the i th sample respectively. $\bar{X}$ and $\bar{Y}$ are the mean of n samples of X_i and Y_i respectively. Here, X represents the three features and Y represents SOH.

The analysis results are shown in Fig. 6:

In Fig. 6, time periods 1, 2, and 3 correspond to the time of extracting F_1 , F_2 , F_3 , respectively. r represents the absolute value of the correlation between the average voltage features extracted for the three time periods and the SOH of the battery. The closer r is to 1, the higher the degree of correlation. The closer r is to 0, the lower the degree of correlation.

The average voltage features of the three selected short periods are highly correlated with the SOH. In comparison, the correlation of the 3

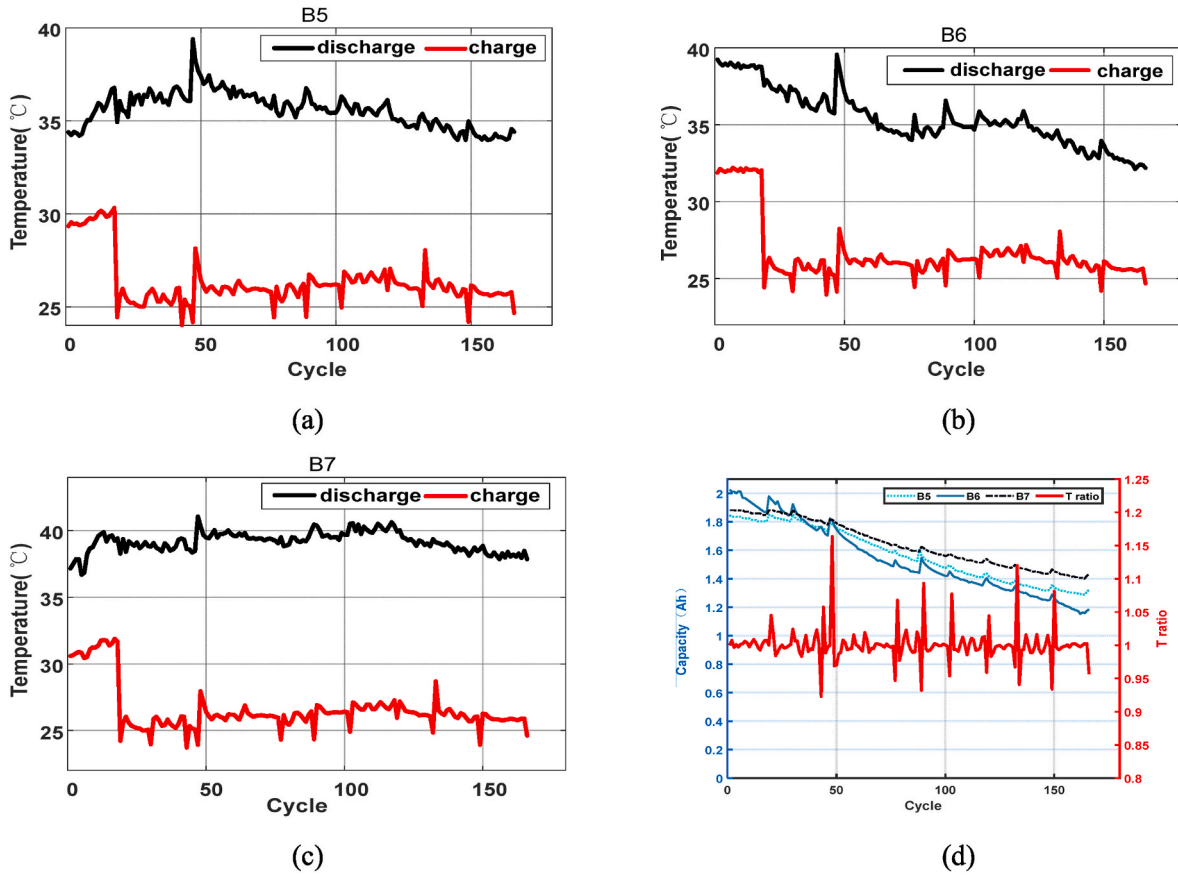


Fig. 8. (a) B5 (b) B6 (c) B7 Temperature at the end of discharge and the beginning of charging. (d) Comparative analysis of temperature ratio and SOH.

min is higher than that of the 1 min, and the correlation is similar to that of the 5 min, but the required time is shorter. Therefore, the short period selected in this paper is 3 min.

3.1.2. Feature extraction based on capacity recovery

To improve the impact of the capacity recovery problem on SOH estimation, this paper specifically extracts features for the battery capacity recovery problem. As stated in the previous section, the capacity recovery of the battery is related to the rebalancing of the active material. Rashid et al. [56] pointed out that, compared with the resting time after charging, the resting time after discharge has a more significant impact on battery performance. Thus, considering the effect of resting time on battery capacity recovery, this paper takes the NASA battery as an example and extracts the resting time of the NASA battery from the end of each discharge to the start of the next charge. Two of the resting times were several days long, therefore, in this paper, we set the upper threshold of resting time to 24 h, set the 24 h to 1, and convert the other times to 24 h as shown in Fig. 7(a).

As shown in Fig. 7(b), the capacity recovery point in the capacity degradation curve of the three batteries corresponds to the fluctuation point of the resting time curve. Therefore, in this paper, the resting time after the end of battery discharge is used as a feature to cope with the recovery of battery capacity.

In addition, temperature, as an external factor, affects the electrochemical reaction inside the battery, and the influence on the capacity and internal resistance of the battery is not negligible [57]. Take NASA batteries as an example. Under constant ambient temperature, the temperatures of B5, B6, and B7 batteries at the end of discharge and the beginning of charging are shown in Fig. 8.

As can be seen from Fig. 8, after the battery is discharged, the temperature of the battery is significantly higher than the ambient

temperature. Hence, the temperature at the beginning of the battery charge is related to the resting time after discharge. Too long resting time leads to the battery temperature being almost the same as the ambient temperature and the temperature tends to be stable, which is consistent with the setting of the upper limit of the resting time. Therefore, this paper chooses to add the initial temperature before battery charging to deal with battery capacity recovery. However, the temperature of the battery after the end of discharge is different, and the temperature before each cycle of charging is also different due to the different resting times. Thus, the surface temperature of the battery cannot be simply regarded as a feature. To highlight the difference in temperature features before each battery charge, this paper chooses the form of temperature ratio to express. The specific approach is: the temperature ratio on the first charge is set to 1. After that, the temperature at the beginning of each cycle charge is compared with the temperature at the beginning of the previous charge to get the ratio of temperature. Among them, the temperature ratio is also set to 1 because the 19th and 47th cycles are rested for too long. Finally, the obtained temperature ratio features were compared with SOH, as shown in Fig. 8 (d).

In Fig. 8(d), there is an obvious temperature ratio peak at the point where the SOH capacity is recovered. This change is similar to the resting time feature. Therefore, the feature of temperature ratio is another feature to cope with the capacity recovery in this paper. For subsequent forecasting, to solve the problem of universality in SOH estimation of different battery types and different operating conditions, the two features proposed in this paper can be used separately or in combination in different situations.

In summary, the 7 features extracted in this paper are as follows.

F_1 : Average charge voltage in 0–3min after 1.5min of battery charge.

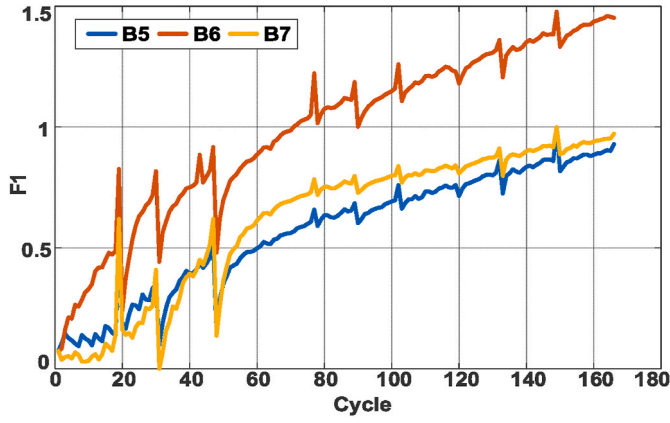


Fig. 9. Feature processing.

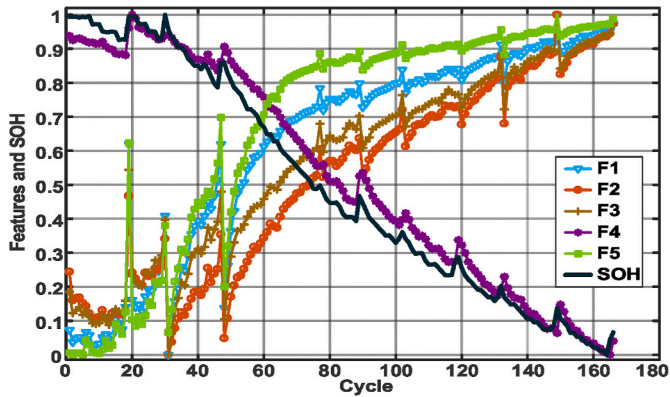


Fig. 10. Extracted features and SOH of B7.

F_2 : Average charge voltage in 3–6min after 1.5min of battery charge.
 F_3 : Average charge voltage in 1.5–4.5min after 1.5min of battery charge.

F_4 : The charging time from 3.8V to 4.1V of the battery.

F_5 : The battery voltage at 1.5 min of battery constant current charging

F_6 : The resting time between the end of battery discharge and the start of the next charge.

F_7 : The ratio of the temperature at the beginning of each battery charge to the temperature at the beginning of the previous charge.

3.1.3. Correlation analysis

The first five feature starting values were uniformly processed. The same feature starting values of B5, B6, and B7 batteries were extracted, and then the minimum of the three starting values was selected as the benchmark, and the other two starting values were reduced to the minimum base value. The remaining data of the two groups of batteries were also reduced by the same reduction multiple of their corresponding starting values.

To facilitate the display of distribution differences among features, after the starting values are unified, the normalized maximum and minimum values of B7 batteries are taken as the benchmark, and the corresponding features of other batteries are also normalized according to Eq. (8). The result of F_1 after processing is shown in Fig. 9.

$$x^* = \frac{x - x_{\min}}{x_{\max} - x_{\min}} \quad (8)$$

The first five features of the B7 battery after treatment were compared with its normalized SOH, as shown in Fig. 10 below.

Compared with the SOH curve of the battery, it can be seen that F_1 ,

Table 2

Correlation analysis between the first five features and SOH.

Battery	F_1	F_2	F_3	F_4	F_5
B5	−0.9537	−0.9852	−0.9815	0.9964	−0.8941
B6	−0.9725	−0.9885	−0.9863	0.9928	−0.9373
B7	−0.9338	−0.9857	−0.9724	0.9916	−0.9049
Cell 1	−0.9921	−0.9900	−0.9892	0.9990	−0.9932
Cell 2	−0.9899	−0.9825	−0.9845	0.9948	−0.9914
Cell 3	−0.9942	−0.9923	−0.9923	0.9992	−0.9924
Cell 4	−0.9935	−0.9933	−0.9926	0.9976	−0.9926
Cell 5	−0.9925	−0.9597	−0.9770	0.9989	−0.9946
Cell 6	−0.9942	−0.9938	−0.9938	0.9961	−0.9924
Cell 7	−0.9949	−0.9910	−0.9925	0.9986	−0.9925
Cell 8	−0.9925	−0.9903	−0.9899	0.9986	−0.9910
CS2_35	−0.9869	−0.9844	−0.9843	0.9979	−0.9843
CS2_36	−0.9911	−0.9819	−0.9860	0.9967	−0.9829

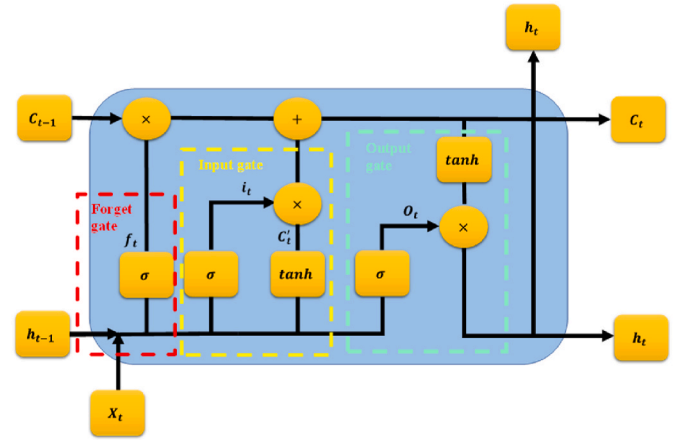


Fig. 11. The structure of LSTM.

F_2 , F_3 , and F_5 are negatively correlated with the SOH of the B7 battery, and F_4 is positively correlated with SOH. The Pearson correlation coefficient is used to calculate the correlation degree.

The correlation degree obtained by combining Eqs. (5)–(7) is shown in Table 2. It can be seen from the table that the voltage features proposed in this paper are highly correlated with SOH, which can be used to reflect the aging information of the battery.

3.2. Proposed the structure of the neural network

The aging of the battery will go through hundreds of cycles, and the capacity degradation information between each cycle is closely related. Therefore, it is crucial to build a model that can learn the long-term correlation information hidden in the battery degradation data. In order to deal with this challenge, the LSTM neural network model is selected in this paper. As shown in Fig. 11, LSTM is mainly composed of three layers of gating neural units, namely the forget gate, input gate, and output gate.

Forget gate: The first layer of LSTM is the forget gate, which decides whether to discard information or not. Whether to discard information or not is determined by the sigmoid(σ) function of the forget gate. The forget gate reads the output h_{t-1} of the previous state of the cell and the input x_t of the current state of the cell, times the weight W_f of the forget gate and adds the bias b_f , and finally passes the sigmoid function Output a value f_t between 0 and 1. If f_t is 0, the information is completely discarded, and if f_t is 1, the information is completely received.

$$f_t = \sigma(W_f \bullet [h_{t-1}, X_t] + b_f) \quad (9)$$

Input gate: After passing the forget gate, the next step is to determine what information is stored in the storage unit and to update the

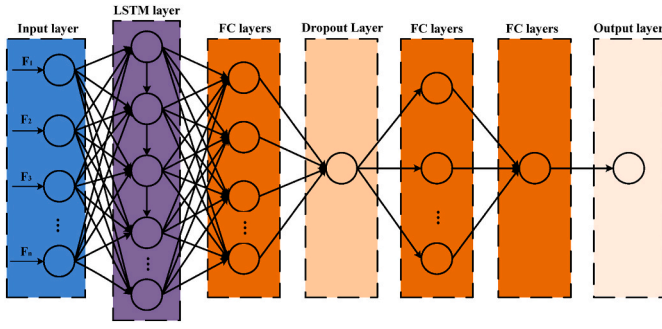


Fig. 12. Structure of neural network in this paper.

information. The input gate is mainly composed of two parts, one is the sigmoid function layer, which decides what information should be updated, and the other is the $\tanh$ function layer, which builds the candidate basis vectors.

$$\begin{cases} i_t = \sigma(W_i \bullet [h_{t-1}, X_t] + b_i) \\ C'_t = \tanh(W_c \bullet [h_{t-1}, X_t] + b_c) \end{cases} \quad (10)$$

where W_i and W_c represent the weight matrix of the sigmoid layer and the weight matrix of the $\tanh$ layer of the input gate. b_i and b_c represent the bias of the input sigmoid and the $\tanh$ layers, so the cell status can be updated in the following way:

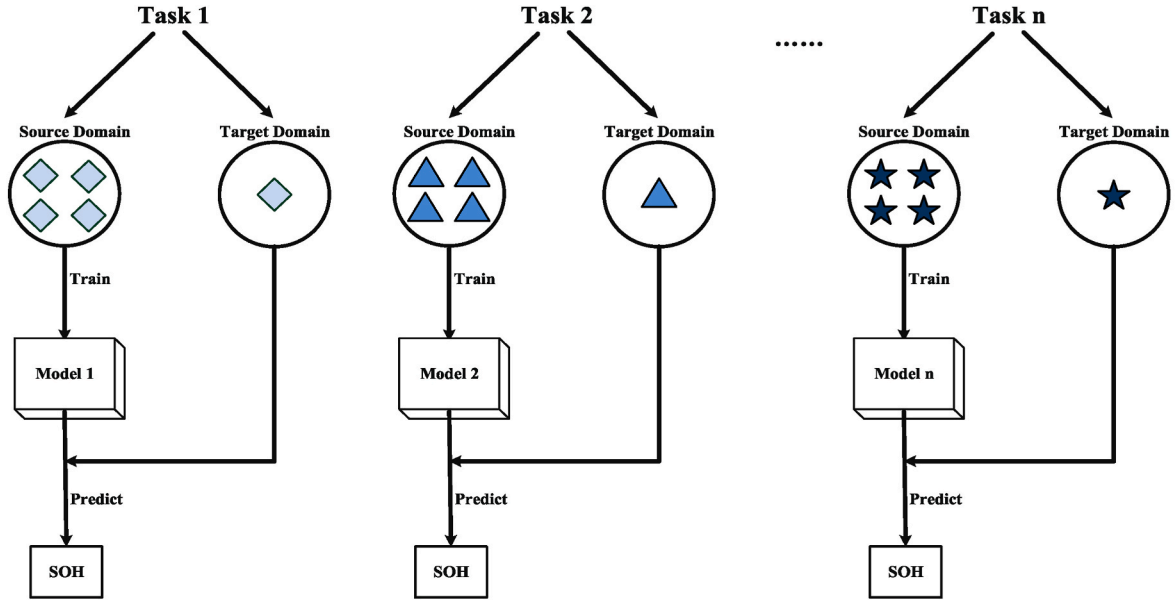
$$C_t = f_t * C_{t-1} + i_t * C'_t \quad (11)$$

where, $C_{(t-1)}$ and C_t represent the previous cell state information and the updated cell state information respectively. C'_t denotes the basis vectors generated by the $\tanh$ function layer, and f_t and i_t both denote numbers between 0 and 1.

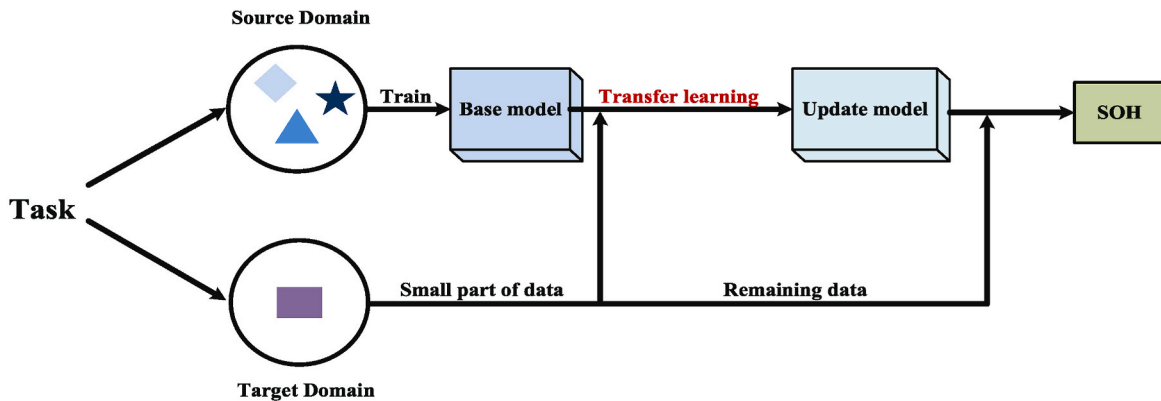
Output gate: Finally, the output gate determines the output of the whole system. The output gate also consists of two parts, one is the sigmoid function layer, which determines what parts of the cell state we want to output. The other part is the $\tanh$ function layer, which performs the $\tanh$ operation on the updated cell state, and the final output result is obtained by multiplying the output of the sigmoid function layer:

$$\begin{cases} O_t = \sigma(W_o \bullet [h_{t-1}, X_t] + b_o) \\ h_t = O_t * \tanh(C_t) \end{cases} \quad (12)$$

where, W_o and b_o represent the weight matrix and bias matrix of the sigmoid layer.



(a)



(b)

Fig. 13. (a) Traditional machine learning. (b) Transfer learning.

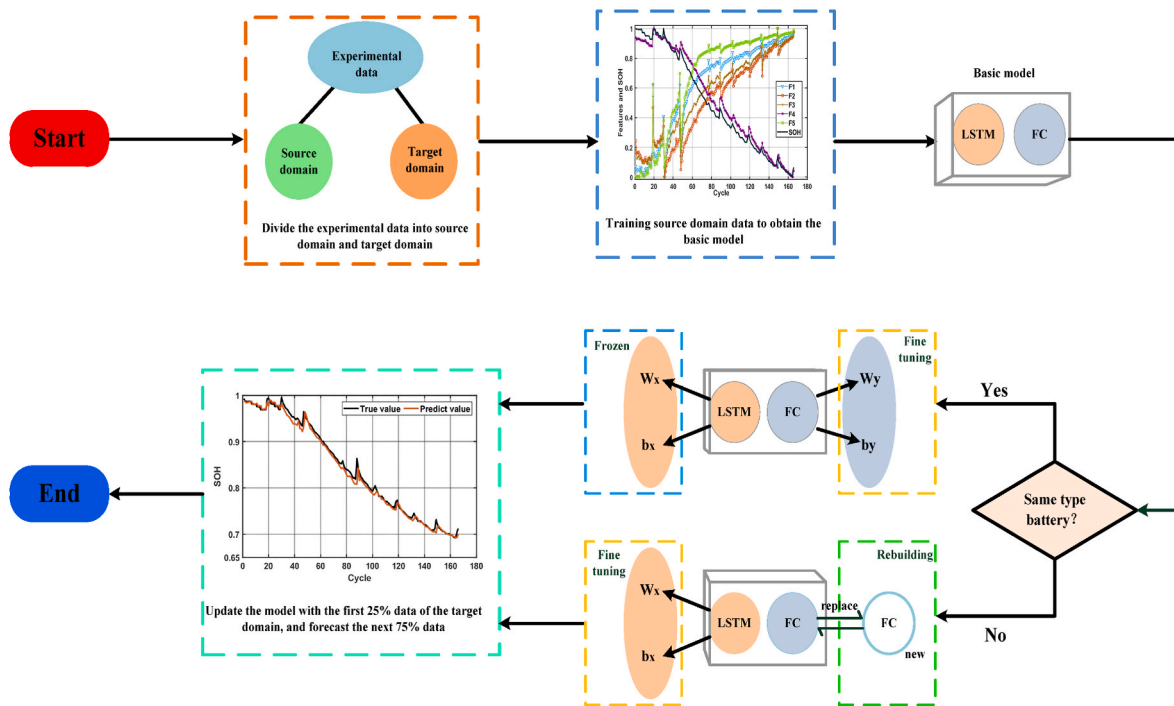


Fig. 14. Two different prediction methods.

Considering that the aging distribution of batteries is complex and different batteries have different degradation processes, in order to effectively learn the relationship between battery health features and SOH, as well as facilitate the transfer learning, the following neural network model structure is established in this paper. As shown in Fig. 12.

The network architecture consists of an LSTM network layer, a dropout network layer and three full connection(FC) layers. Among them, the LSTM network layer and Dropout layer can prevent gradient disappearance, gradient explosion, and overfitting problems that occur in the model during learning [48,58]. As a result, it can enhance the generalization ability of the model. The addition of the full connection layer can make the model learn the specific properties of the battery in the target domain better, and realize the accurate transfer learning of the model.

3.3. Proposed SOH prediction method based on transfer learning

In terms of SOH prediction, traditional machine learning methods are to train specific models based on specific data, as shown in Fig. 13 (a). If the distribution of the data changes, the model has to be retrained. In this way, every time a model is trained, the corresponding data must be collected first, which not only wastes time but also requires expensive costs. Compared with traditional machine learning methods, transfer learning is more in line with the idea of using knowledge across tasks in practical applications, as shown in Fig. 13(b). In addition, since battery decay is similar, transfer learning is able to take advantage of this and build a basic model to understand the general knowledge of battery degradation. Then, different prediction methods (fine-tuning or rebuilding) are adopted for different prediction situations, and a small amount of target domain data is combined to update model parameters to achieve an accurate estimation of target domain SOH.

The proposed SOH prediction method in this article is detailed in Fig. 14. At first, the experimental data is divided into two categories, one is source domain data: used to train the basic model. The other is the target domain data: used to update the basic model for subsequent predictions.

After classification, extract the same features in the source and target domains, respectively. The features in the source domain are used to train the basic model. Then, according to whether the target domain battery data is the same type of battery data as the source domain battery data. Update the model using the fine-tuning or rebuilding strategy, respectively.

(1) Transfer learning strategy for the same type of batteries

Fine-tuning is aimed at the same type of battery. Because the LSTM network can learn long-term correlations between data, and the degradation information of the same type of battery is more similar. Therefore, after training the basic model, the parameters of the LSTM layer remain unchanged, only the parameter learning rate of the fully connected layer is reduced, and then the weight and bias parameters of the fully connected layer are updated according to the first 25% data of the battery in the target domain. Consequently, the fine-tuning SOH estimation model could work.

(2) Transfer learning strategies for different types of batteries

Rebuilding is aimed at different types of batteries, and the degradation information of different types of batteries is too different. However, to retain the common information of battery degradation, after training the basic model, the learning rate of the LSTM model is reduced, and the weights and bias parameters are fine-tuned. At the same time, The fully connected layer is rebuilt in order to better learn the characteristics of the new battery. The model parameters are then updated based on the top 25% of the target domain battery data, and the rebuilding prediction of SOH is realized.

4. Results and discussion

4.1. The evaluation criteria

In this section, the performance of the proposed method will be verified from the following three aspects.

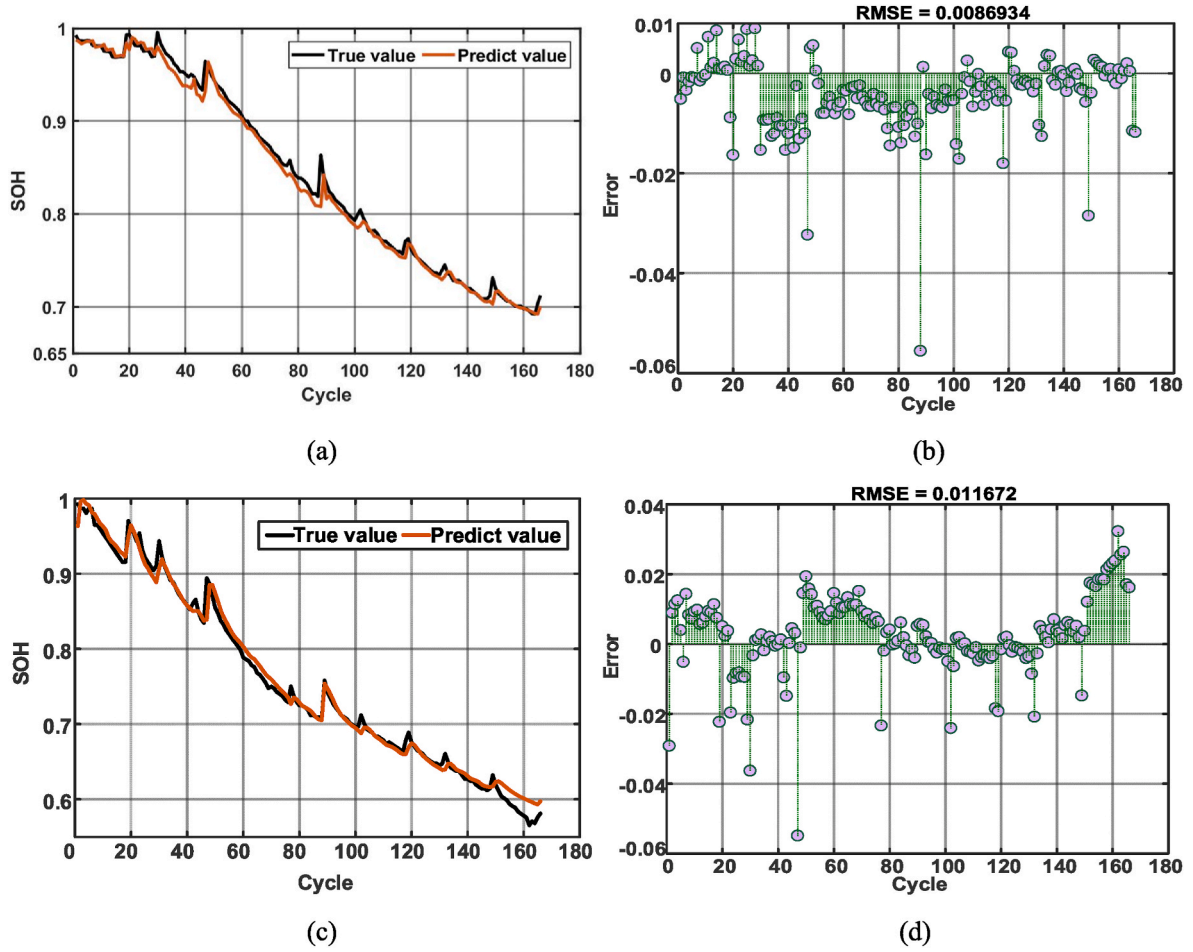


Fig. 15. Transfer learning prediction and error results for the same type batteries: (a) prediction result of B5; (b) error result of B5; (c) prediction result of B6; (d) error result of B6.

- (1) The performance of the features presented in this paper.
- (2) Estimation effect of transfer learning in this paper.
- (3) Comparative analysis results with BPNN, SVR, and GPR.

This paper uses root mean square error (RMSE), mean square error (MSE), and maximum absolute error (MAX) to express the evaluation effect of prediction. The formula for the three errors is as follows:

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (13)$$

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (14)$$

$$MAX = \max |y_i - \hat{y}_i| \quad (15)$$

where n represents the number of predicted battery data samples. y_i represents the true SOH of the battery per cycle, and $\hat{y}_i$ represents the estimated SOH of the battery per cycle.

4.2. The performance of the features presented in this paper

To prove the estimation effect of the features proposed in this paper, the fine-tuning transfer learning strategy proposed in this paper is used for verification. In addition, BPNN, SVR, and GPR are tested with the same training set and test set. On the one hand, it is used to verify the effectiveness of features. On the other hand, it is compared with the fine-tuning strategy proposed in this paper to show the effectiveness of the

fine-tuning strategy proposed in this paper.

The transfer learning training strategy for the same type of batteries is as follows: Select a group of batteries as the target domain, and other batteries of the same type as the source domain. The battery data in the source domain is used to train the basic model. The first 25% of data in the target domain is used to update the basic model, and the remaining 75% of SOH is predicted. The predictions are as follows.

4.2.1. Results of the NASA dataset

The B5 and B6 batteries are also from the NASA battery data set and have the same number of samples as the B7 batteries. Before the basic model is updated, the LSTM layer weights and bias parameters of the trained basic model are frozen and kept unchanged. Only the weight and bias learning rate of the FC layer are reduced, and then the fine-tuning strategy is implemented to update the basic model.

Fig. 15(a)-(d) shows the predicted SOH and error results of B5 and B6 batteries. It can be found that the transfer learning performance of B5 is better than that of B6 batteries (RMSE of B5 = 0.00870, RMSE of B6 = 0.01170), which is reasonable. As can be seen from Fig. 3, the downward trend of SOH of the B5 battery is the same as that of the B7 battery, while the downward trend of SOH of the B6 battery is quite different from that of the B5 battery and B7 battery. Therefore, When the source domains are B6 and B7 batteries, the training base model can learn the downward trend of the B7 battery, and the task knowledge has good applicability for predicting the B5 battery. On the contrary, for the basic model whose source domain is B5 and B7 batteries, the applicability of task knowledge to B6 batteries is relatively poor. This is the same reason as the prediction performance of the B7 battery based on B5 and B6 as the

Table 3

Prediction results of the same type of batteries.

Training	Test	RMSE (%)	MSE (%)	MAX (%)
B6, B7	B5	0.87	0.76	5.5
B5, B7	B6	1.17	1.37	5.5
B5, B6	B7	0.87	0.76	5.5
Cs2_36	Cs2_35	1.05	1.10	2.52
Cs2_35	Cs2_36	1.02	1.04	3.73
Cell 2-8	Cell 1	0.55	0.30	1.17
Cell 1, Cell 3-8	Cell 2	1.20	1.44	4.82
Cell 1-2, Cell 4-8	Cell 3	0.48	0.23	1.32
Cell 1-3, Cell 5-8	Cell 4	0.58	0.34	1.06
Cell 1-4, Cell 6-8	Cell 5	1.33	1.76	7.79
Cell 1-5, Cell 7-8	Cell 6	0.61	0.37	1.08
Cell 1-6, Cell 8	Cell 7	0.46	0.21	1.17
Cell 1-7	Cell 8	0.59	0.35	1.72

source domain data in Table 3 below is higher than that of the B6 battery based on the basic model trained on B5 and B7 as the source domain.

4.2.2. Results of the Oxford dataset

In Fig. 16, the number of samples in the two battery datasets of the Oxford battery is different in size, and there is also a local SOH plunge in the Cell 2 and Cell 5 data sets. Therefore, in terms of the prediction effect of transfer learning, the performance of Cell 2 and Cell 5 cells is significantly worse than that of the other 6 cells (RMSE is below 0.61% except for Cell 2 and Cell 5 cells which are above 1%, and MAX is below 1.72% except for Cell 2 and Cell 5 cells which are above 4%). However, since the SOH of all eight Oxford batteries decreased in the same way, the source-domain task knowledge was well transferred to the new model. Therefore, both from the perspective of RMSE and MAX, the accuracy and stability of the prediction results can be well guaranteed.

4.2.3. Results of the CALCE dataset

In Fig. 17, CS2-35 and CS2-36, are from the CALCE battery dataset with the same number of samples. This paper uses the first 800 cycles of data from two datasets for training predictions. As can be seen from Fig. 3, the SOH decline trend of CS2-35 and CS2-36 cells shows constant fluctuation in a small range. However, due to the same task knowledge of small range fluctuation of data samples in the source domain and target domain samples, the prediction performance after transfer learning can obtain high accuracy. It can be seen from the predicted MAX of the two cells (the MAX of CS2-35 is below 3.73% and the MAX of CS2-36 is below 2.52%) that the prediction results have high stationarity and can well track the SOH changes in a very small range.

Table 3 shows the results of all same-type batteries in the form of RMSE, MSE, and MAX respectively.

4.2.4. Comparison of other methods for the same type of battery

To verify the prediction effect of the proposed features on SOH, the proposed method is compared with BPNN, SVR, and GPR, using the same proportion of training set and test set and the same experimental features. Fig. 18 illustrates the prediction between the same type of batteries, with the B5 battery as the target domain. The results of the different methods are shown in Table 4.

The precision of the model is related to features and the model itself. As can be seen from Fig. 18 and Table 4, except for the NASA dataset, SVR, BPNN and GPR models can all obtain small SOH estimation errors, indicating that the health features proposed in the paper can effectively characterize the battery degradation. Moreover, compared with the Oxford dataset and the CALCE dataset, the batteries in NASA dataset are more inconsistent, and the distribution of the training set and test set is more different, which results in the larger estimation errors of SVR, BPNN and GPR models. Due to the fine-tuning strategy, the model proposed in this paper can adjust the base model parameters according

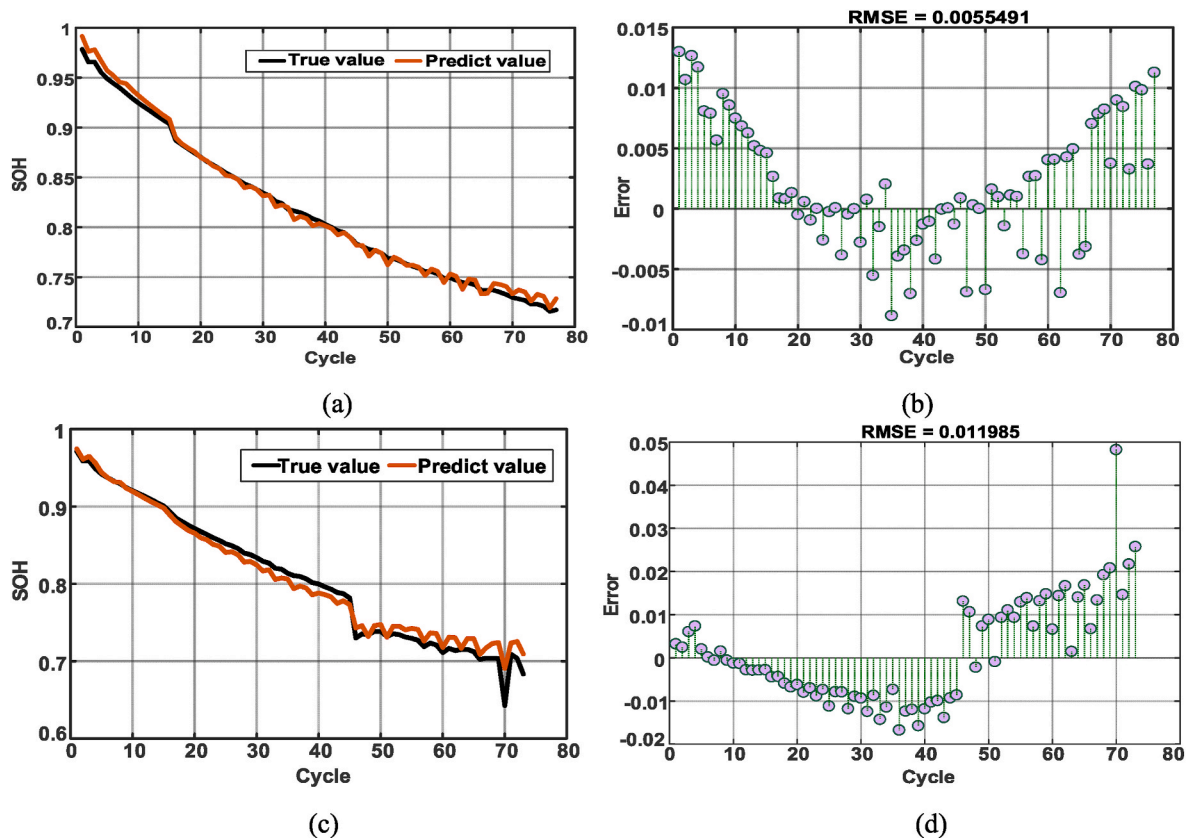


Fig. 16. Transfer learning prediction and error results for the same type batteries: (a) prediction result of Cell 1. (b) error result of Cell 1. (c) prediction result of Cell 2. (d) error result of Cell 2.

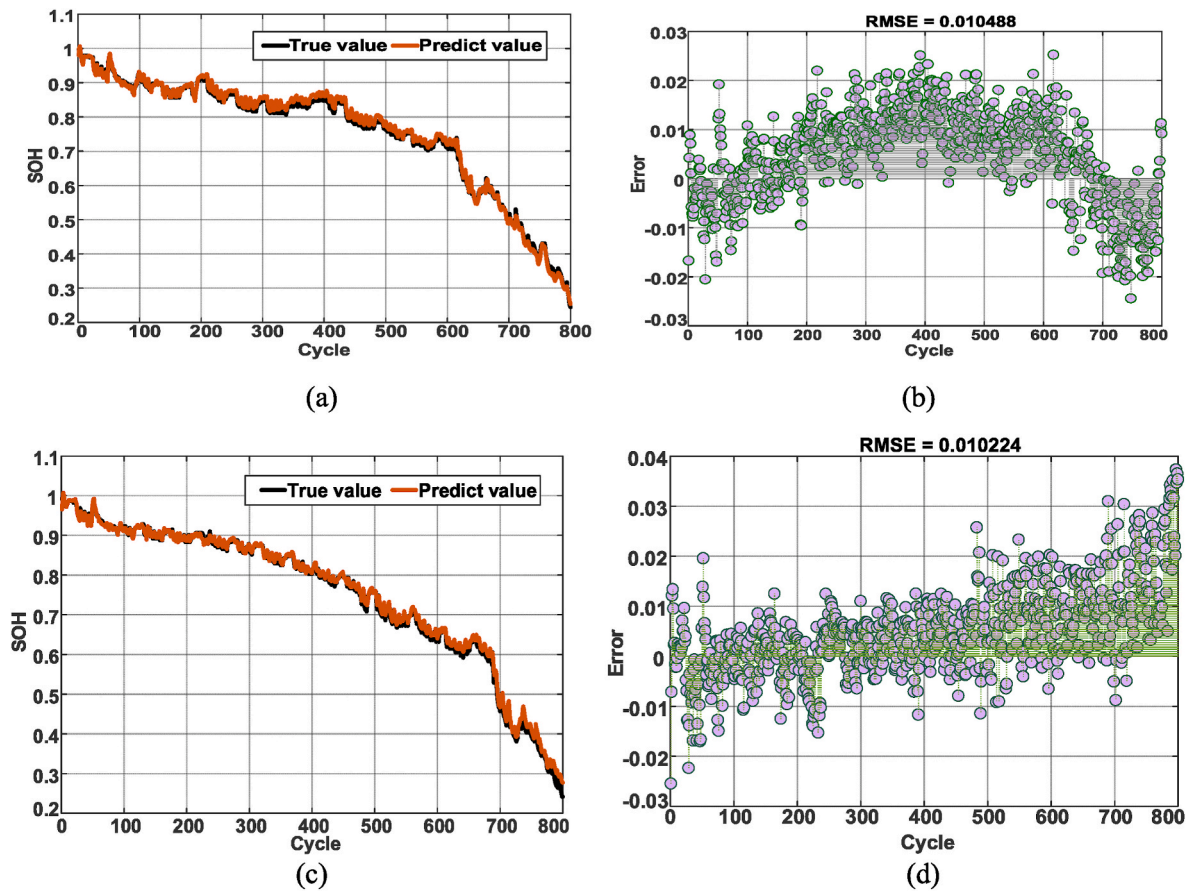


Fig. 17. Transfer learning prediction and error results for the same type batteries: (a) prediction result of CS2-35. (b) error result of CS2-35. (c) prediction result of CS2-36. (d) error result of CS2-36.

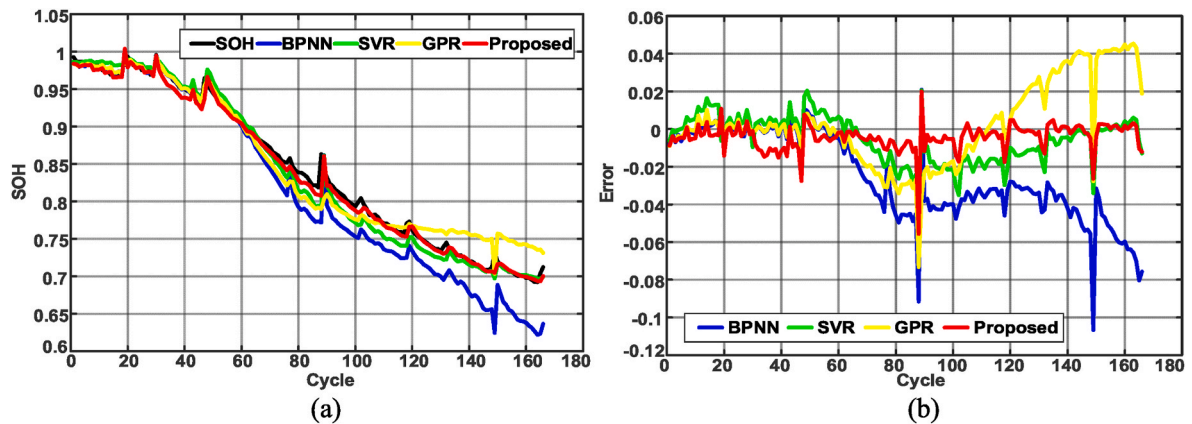


Fig. 18. Comparative analysis.

to the target battery to make it more suitable for its own characteristics, so it can obtain SOH estimation results with high precision.

4.3. Estimation effect of transfer learning in this paper

To prove the performance of the proposed transfer learning strategy, the prediction effect with transfer learning (LSTM-Transfer) was compared with that without transfer learning (LSTM). The prediction results of transfer learning are also compared with other methods.

The transfer learning strategy among different types of batteries is as follows: Select one type of battery data as the source domain, and the

other type as the target domain. The battery data in the source domain is used to train the basic model. The first 25% of data in the target domain is used to update the basic model, and the remaining 75% of SOH is predicted. The predictions are as follows.

4.3.1. Transfer learning of NASA and CALCE

In training the base model, as in the former section, the source domain data are used for training. When updating the model, the fully connected layer of the trained basic model is discarded, the new fully connected layer is reset, and the learning rate of the weight and bias parameters of the LSTM layer is reduced. Thus, the fine-tuning of the

Table 4
Experimental results of same type batteries.

Training	Test	Evaluation criterion	BPNN [9]	SVR [38]	GPR [40]	Proposed method
B6, B7	B5	RMSE (%)	4.03	1.95	2.57	0.87
B5, B7	B6	RMSE (%)	3.77	4.21	6.93	1.17
B5, B6	B7	RMSE (%)	1.83	1.01	0.90	0.87
Cs2_36	Cs2_35	RMSE (%)	1.30	1.14	1.11	1.05
Cs2_35	Cs2_36	RMSE (%)	1.69	1.15	1.43	1.02
Cell 1-8	Cell 1	RMSE (%)	0.83	0.93	1.10	0.55
Cell 1, Cell 3-8	Cell 2	RMSE (%)	2.08	0.85	3.47	1.20
Cell 1-2, Cell 4-8	Cell 3	RMSE (%)	0.77	0.41	0.56	0.48
Cell 1-3, Cell 5-8	Cell 4	RMSE (%)	1.23	1.44	1.65	0.58
Cell 1-4, Cell 6-8	Cell 5	RMSE (%)	1.66	1.35	1.23	1.33
Cell 1-5, Cell 7-8	Cell 6	RMSE (%)	0.58	0.44	1.34	0.61
Cell 1-6, Cell 8	Cell 7	RMSE (%)	1.00	1.59	1.40	0.46
Cell 1-7	Cell 8	RMSE (%)	0.91	1.22	1.40	0.59

LSTM layer and the rebuilding of the fully connected layer are realized.

Fig. 19(a)–(b) shows the prediction results of NASA battery as the source domain, CALCE battery as the target domain, and Fig. 19(c)–(d) show the CALCE battery as the source domain, NASA battery as the target domain. The experimental results show that the transfer learning method used in this paper can learn the common knowledge of different types of batteries well and achieve stable prediction performance. The

prediction curves of the whole process almost overlap with the actual values. In contrast, the prediction result without using the transfer learning strategy is obviously too much error, which will not be applied to the prediction between different types of batteries.

4.3.2. Transfer learning of NASA and Oxford

The mutual transfer learning results between NASA and Oxford batteries can also track the actual SOH descent curve well, which can be seen in Fig. 20(a)–(d), and Table 5. The RMSE of the NASA battery is below 1.44% and the MAX is below 4.8%. RMSE of Oxford battery is below 1.27% and MAX is below 4.63%.

As can be seen in both Fig. 20, the estimation results with transfer learning are significantly better than the estimation results without the use of transfer learning. Therefore, it can be concluded that it is advantageous to use the rebuilding transfer learning strategy proposed in this paper at the time of different kinds of battery prediction.

Subsequently, Table 5 shows the transfer learning of all different types of batteries in the form of RMSE, MSE, and MAX respectively.

4.3.3. Comparison of other methods for the different types of battery

As can be seen from the RMSE in Table 6, the transfer learning method proposed in this paper has obvious advantages in the prediction between different batteries, and the estimated accuracy of RMSE is significantly higher than that of the other three methods. The other three methods have a great deviation from the real value and lose the accuracy of estimation. It can be concluded that the degradation distribution of different batteries is too different, and the parameters of the basic model are difficult to adapt to the new battery data. Therefore, fine-tuning the LSTM layer and rebuilding the new fully connected layer can better

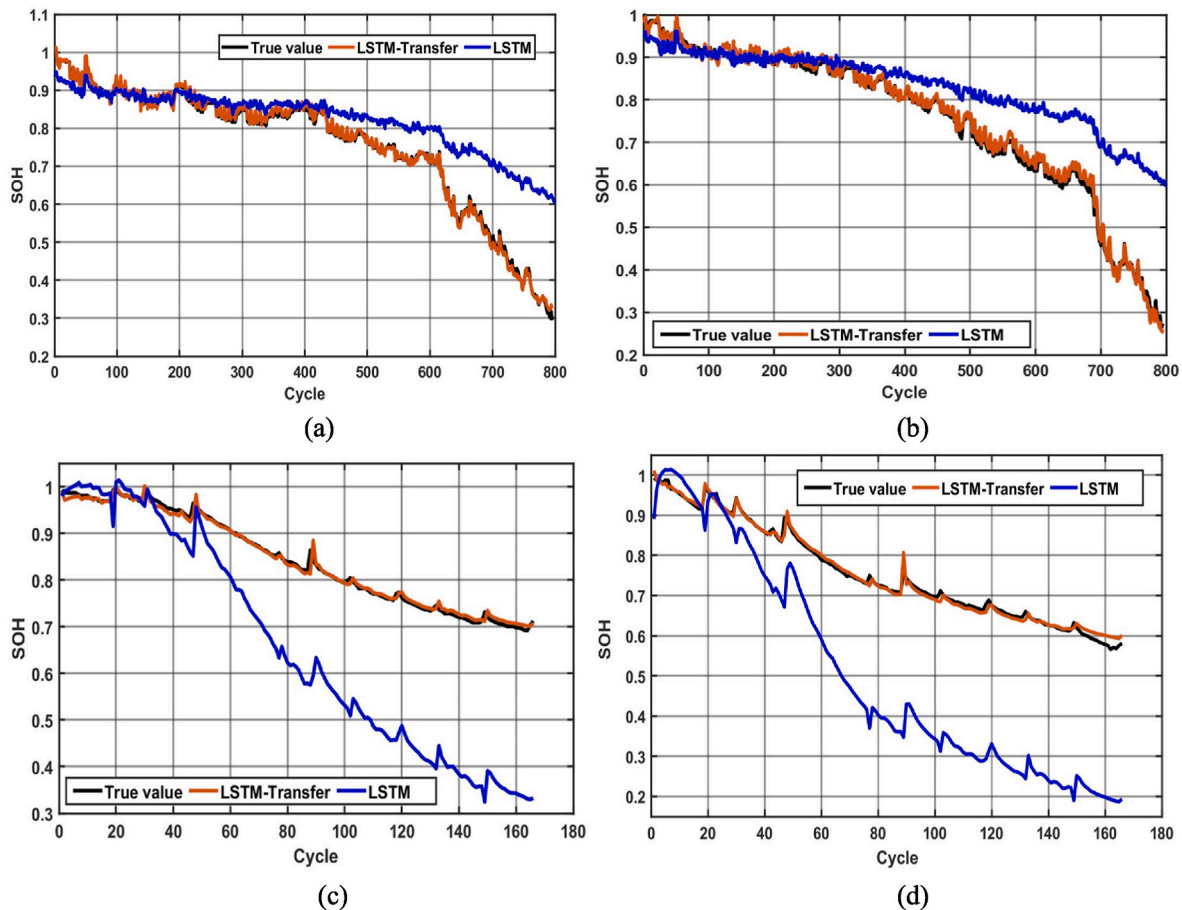


Fig. 19. Transfer learning prediction results for the different batteries: (a) prediction result of CS2-35. (b) prediction result of CS2-36. (c) prediction result of B5. (d) prediction result of B6.

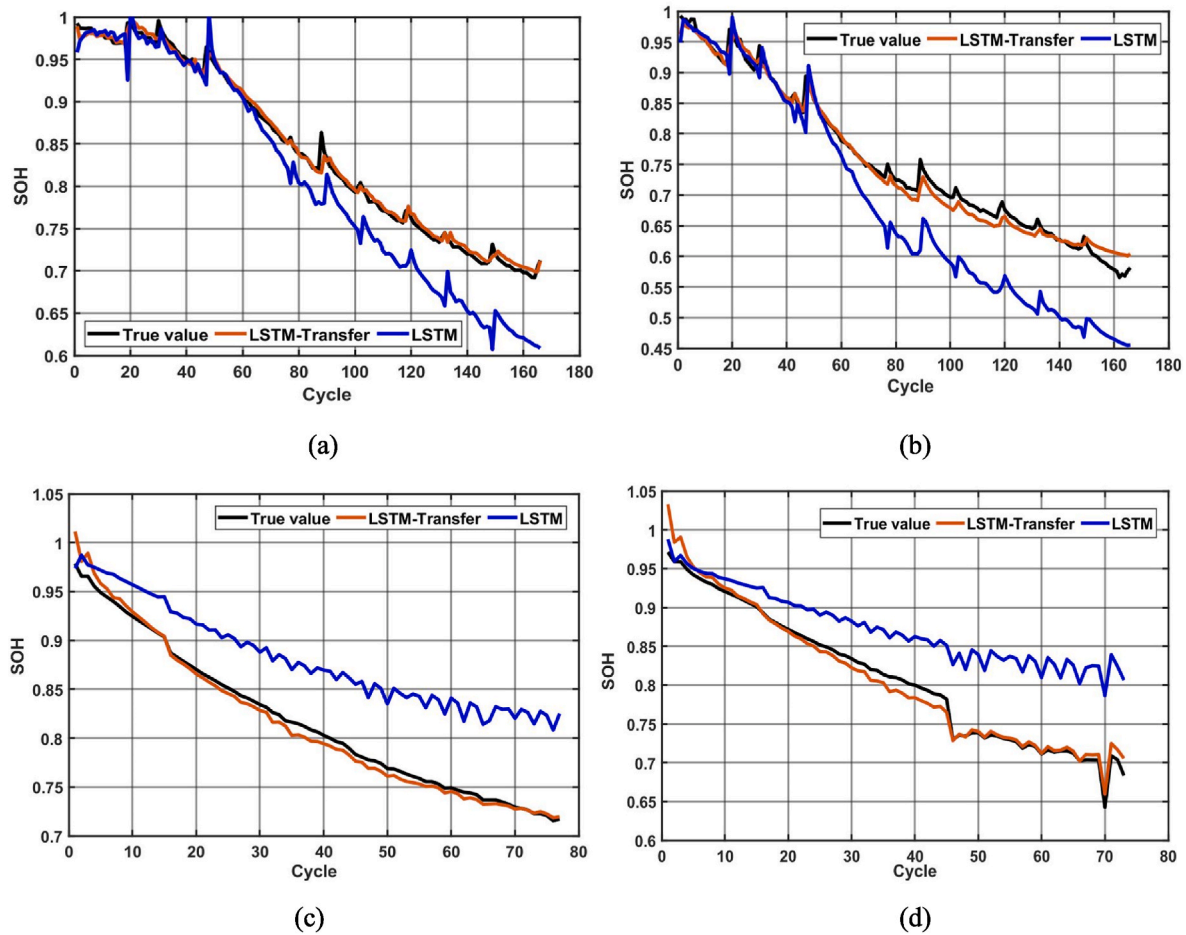


Fig. 20. Transfer learning prediction results for the different batteries: (a) prediction result of B5. (b) prediction result of B6. (c) prediction result of Cell 1. (d) prediction result of Cell 2.

Table 5

Prediction results of different types of batteries.

Training	Test	RMSE (%)	MSE (%)	MAX (%)
B5, B6, B7	Cs2_35	0.94	0.88	3.18
B5, B6, B7	Cs2_36	1.07	1.14	3.46
CS2-35, CS2-36	B5	0.86	0.74	5.00
CS2-35, CS2-36	B6	1.09	1.19	5.00
CS2-35, CS2-36	B7	0.91	0.82	1.93
B5, B6, B7	Cell 1	0.79	0.63	1.38
B5, B6, B7	Cell 2	1.27	1.61	2.21
B5, B6, B7	Cell 3	0.68	0.46	1.39
B5, B6, B7	Cell 4	0.77	0.59	0.78
B5, B6, B7	Cell 5	1.12	1.25	4.63
B5, B6, B7	Cell 6	0.67	0.45	1.38
B5, B6, B7	Cell 7	0.77	0.59	1.43
B5, B6, B7	Cell 8	0.75	0.56	1.50
Cell 1-8	B5	0.69	0.48	4.73
Cell 1-8	B6	1.44	2.07	4.42
Cell 1-8	B7	1.25	1.56	4.80

realize the learning of the target domain data and update the model parameters, so as to achieve accurate transfer learning prediction of the model.

5. Conclusions

SOH is an important indicator of battery performance. To improve SOH estimation precious and solve the problem that the established SOH model is difficult to suit for different types of batteries or even the same

Table 6

Experimental results of different type batteries.

Training	Test	Evaluation criterion	BPNN [9]	SVR [38]	GPR [40]	Proposed method
B5, B6, B7	CS2-35	RMSE (%)	19.06	14.19	25.56	0.94
	CS2-36	RMSE (%)	13.08	15.36	26.76	1.07
	Cell 1	RMSE (%)	4.30	17.20	10.36	0.79
	Cell 2	RMSE (%)	5.12	18.03	13.71	1.27
	Cell 3	RMSE (%)	5.37	16.26	10.22	0.68
	Cell 4	RMSE (%)	4.18	16.80	10.32	0.77
	Cell 5	RMSE (%)	5.84	15.43	15.23	1.12
	Cell 6	RMSE (%)	4.79	14.67	9.17	0.67
CS2-35, CS2-36	Cell 7	RMSE (%)	2.36	13.74	8.07	0.77
	Cell 8	RMSE (%)	3.48	15.61	11.10	0.75
	B5	RMSE (%)	16.70	13.35	20.85	0.86
	B6	RMSE (%)	13.16	14.20	14.00	1.40
Cell 1-8	B7	RMSE (%)	12.01	7.53	11.40	0.87
	B5	RMSE (%)	7.20	10.33	14.65	0.69
	B6	RMSE (%)	8.38	5.36	13.84	1.44
	B7	RMSE (%)	5.48	3.31	14.74	1.25

type of batteries due to the inherent cell inconsistency, this paper presents a SOH estimation framework based on new feature mining and transfer learning.

The main merits of the proposed framework can be summarized as follows: (1) the five voltage health features and the two capacity recovery features can well characterize the battery degradation. Moreover, these features are obtained from partial constant current charging data, compared with some health features which need the whole constant current and constant voltage charging mode, which can reduce the requirement of the feature acquisition restrictions. (2) The neural network model proposed in the paper contains an LSTM layer, a dropout layer and three full connection layers. The model structure has a different division of labor in each layer, which can not only fully learn the mapping relationship between features and SOH as well as the sequential dependence of degradation, moreover, it facilitates different degrees of migration. (3) Battery degradation patterns are similar but different, generally speaking, the same type of battery has a large similarity, but different types have a small similarity. The rebuilding and fine-tuning strategy proposed in this paper can better solve this problem. Compared with the transfer learning of the same battery type, different battery types pay more attention to learning knowledge from the target battery.

When the inconsistency between the training battery and the test battery is small, the SOH estimation model constructed by the features proposed in this paper and the traditional method (BPNN, SVR, and GPR) can obtain better estimation results, indicating that the features proposed in this paper are meaningful. The greater the inconsistency between the training battery and the test battery, the greater the prediction deviation of these methods, which fully demonstrates the necessity of model migration. For the method proposed in the paper, the RMSE remains stable below 1.44% under different conditions regardless of the degree of inconsistency between the training battery and the test battery.

Although the proposed SOH estimation framework can obtain high-precision estimation results, there are still some limitations that need further research in future work. (1) In order to obtain health features, the battery needs to be continuously charged to 4.1V, it is necessary to further reduce the restrictions on charging conditions and charging time to meet the needs of more users in practical applications. (2) For retraining the base model in transfer learning, the adjustment of model parameters needs to be manually set. How to automatically determine the degree of battery inconsistency and set model parameters is the future research work.

Credit author statement

Kai Huang: Conceptualization, Methodology, Funding acquisition, Software, Formal analysis, Writing- Original draft preparation, Writing- Reviewing and Editing. Kaixin Yao: Methodology, Data curation, Software, Formal analysis, Writing- Original draft preparation. Yongfang Guo: Supervision, Data analysis, Software, Visualization, Writing- Reviewing and Editing. Ziteng Lv: Validation, Writing- Reviewing and Editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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